

Asymptotic Flatness and BMS

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Abstract

Notes on asymptotic flatness and the BMS group.

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1 Geometric preliminaries

Let M be a four-dimensional spacetime, *i.e.*, a real smooth 4-manifold endowed with a metric g metric signature $(-, +, +, +)$, plus a smoothly varying choice made at every $T_p M$ as to which time-like and null vectors are future-versus past-directed. Any future-directed null geodesic which doesn't eventually get permanently trapped in an isolated system¹ is said to run off to *future null infinity* $\mathcal{I}^+$. *Past null infinity* $\mathcal{I}^-$ is defined analogously. We will see later that conformal compactifications let us more literally interpret $\mathcal{I}^\pm$ as boundary components.

Let $N \subset M$ be a smooth codimension-1 submanifold. For $p \in N$, a vector $n \in T_p N$ is said to be *normal* to N at p if it is orthogonal to all vectors tangent to N at p , *i.e.*, if $\langle n, m \rangle = 0$ for all $m \in T_p N$. Conversely, given such an n the tangent space $T_p N$ is exactly the set of all $m \in T_p M$ such that $\langle m, n \rangle = 0$. Just as in the Euclidean case, at fixed p and N such vectors form a 1-dimensional vector space. By choosing a normalization convention and a local orientation for N , we may therefore speak of *the* normal vector $n = (n^a)$ or the corresponding normal covector $(n_a) \equiv (g_{ab} n^b) = g(n, \cdot)$.

N is called a *null hypersurface* if, at every point $p \in N$, the² normal vector (n^a) is null, or equivalently if the normal covector (n_a) . Then from our observations above we see that $n \in T_p N$, so that n is simultaneously tangent and normal to N at p . Now we abuse notation slightly and extend n to a vector field defined everywhere on N . Then the integral curves of n are all trajectories within N , and cover the latter. We call these null curves the *null generators* of N . If everything is analytically nice enough, then the set of integral curves (equivalently, the set of all equivalence classes $[p \in N]$ where $p \sim q$ iff some integral curve passes through both points) inherits the structure of a real smooth two-manifold. If this occurs, a smooth family of null generators is called a *null congruence*. This smooth structure may hold in some regions but not in others. A *caustic* is a point where neighboring generators intersect or focus so strongly that the inherited smooth structure breaks down.

Suppose we have covered N by such a null congruence of null generators. A *cut* of N is a smooth space-like³ 2-submanifold of N which intersects each null generator exactly once in some region under consideration. Choosing a cut of a null hypersurface should be thought of as the null-geometry analogue of choosing a spatial

¹This is deliberately loose phrasing which we'll make more precise later

²Here 'the' is a mild abuse of vocabulary since it relies on choices of orientation and normalization, but all such choices agree on whether N is null.

³That is, having exclusively spacelike or vanishing tangent vectors.

slice of a Lorentzian manifold.

Why should cuts exist at all? In more algebraic terms and rephrased slightly, why should the induced metric on N have signature $(0, +, +)$? Recall that, algebraically, the signature of a quadratic form $F : V \times V$ over a real vector space of finite dimensions is defined procedurally:

- Pick any basis for F w.r.t. which the component expression for F is diagonal.
- Read off the number of positive, negative, and vanishing diagonal components; this triple is the signature of F .
- Step 1 is provably possible, and the results of Step 2 are provably independent of choices made during Step 1.

So the claim that N has induced metric signature $(0, +, +)$ (or more precisely, that every tangent space $T_p N$ has inner product with this signature) says that there exists a basis of three vectors in $T_p N$ which are pairwise orthogonal and which are respectively null, spacelike, and spacelike. To show this, we take the first of these vectors to be n itself. By definition $\langle n, m \rangle = 0$ for all $m \in T_p N$, so n is null and any completion to a basis (n, k, k') automatically obeys $\langle n, k \rangle = \langle n, k' \rangle = 0$; the only non-automatic constraints remaining are that (n, k, k') be linearly independent (which excludes only a measure-0 set of sets) and that $\langle k, k' \rangle = 0$ (which can be accomplished by subsequently applying a Gram-Schmidt procedure). What, then, can we say for sure about k, k' ? We know that no null A and timelike B obey $\langle A, B \rangle = 0$, so k, k' must both be null or spacelike. We also know that two nonzero null vectors A, B satisfy $\langle A, B \rangle = 0$ iff one is a rescaling of the other, but k, k' are both assumed linearly independent of n . So by elimination k, k' are both spacelike, and the induced metric on N must have signature $(0, +, +)$.

To build a field theory on M , we define a collection of fields to be fibered over M plus prescribed boundary conditions⁴ The *(on- / off-)shell asymptotic phase space* of our theory is the set of all histories obeying the prescribed boundary conditions (and / but not necessarily) the equations of motion.

2 Bondi Gauge

When we must work in coordinates, it will be convenient to locally work in *Bondi gauge*, which is well-adapted to studying null geodesics. Here we develop the Bondi gauge in slightly different language from the standard references [?, ?, ?]. First, take a null hypersurface N and extend it locally to a 1-parameter family of such surfaces $\{N_u\}_u$ parametrized by a real variable u . This can be done in some region about a point of interest, but attempting to extend it to a smooth foliation $M = \cup_u N_u$ may run into geometric obstructions (in the form of, *e.g.*, caustics) or global topological obstructions, so already our gauge choice is at most guaranteed to work only locally. This gives us our first local coordinate u , telling which copy of N a point lives on.

Let n be a null vector field defined near our point of interest, such that at every point $p \in N_u \subset M$ the local value $n(p) \in T_p M$ is normal and tangent to N_u . From n we can construct null generators as described above, so that by construction every point in our region lies on exactly one generator. Now along each N_u pick a cut $\Sigma_u \subset N_u$, chosen such that N_u varies smoothly with u . Pick coordinates $(x^A)_{A=1,2}$ along each cut Σ_u , then extend these to coordinates on N_u by requiring that they be constant along each null generator; we can locally do so in such a way that x^A varies smoothly with u . This gives us coordinates (u, x^A) .

At this point we have by construction that u and x^A are constant along generators, so we take the remaining coordinate r to grow along generators, and to vary smoothly more generally. Since moving along a particular generator changes r while leaving (u, x^A) fixed, we have automatically that ∂_r points along the null generators, so ∂_r is some local rescaling of n .

What do these geometric choices tell us about the metric in components (u, r, x^A) ? First, since ∂_r points

⁴More precisely [?], in defining a variation problem we should think of spatial and null boundary conditions as defining a theory, but leave future/past boundary conditions undefined as these define states in the theory.

along the null generators, ∂_r must be null, hence

$$g_{rr} = g(\partial_r, \partial_r) = 0 \quad (1)$$

Next, varying x^A at fixed (u, r) keeps us in the same N_u , so ∂_A is tangent to N_u . But $n = \partial_r$ is normal to N_u , hence by definition orthogonal to every vector tangent to N_u . So we have

$$g_{rA} = g(\partial_r, \partial_A) = 0. \quad (2)$$

Our metric thus takes the form

$$ds^2 = g_{uu}du^2 + 2g_{ur}dudr + 2g_{uA}dudx^A + g_{AB}dx^A dx^B. \quad (3)$$

We write this coordinate system as $(x^a) = (u, r, x^A)$. For convenience we now rename our coefficients. First, from our earlier discussion of cuts, metric signature considerations ensure that (g_{AB}) is strictly positive-definite, hence invertible as an array. We therefore define

$$U^A \equiv g^{AB}g_{uB} \implies g_{uA} = g_{AB}U^B. \quad (4)$$

Next we complete the square and further rename as

$$\begin{aligned} ds^2 &= g_{uu}du^2 + 2g_{ur}dudr + 2g_{AB}U^Bdudx^A + g_{AB}dx^A dx^B \\ &= (g_{uu} - g_{AB}U^A U^B) du^2 + 2g_{ur}dudr + g_{AB}(U^A du + dx^A)(U^B du + dx^B) \\ &\equiv -Fdu^2 - 2e^\beta dudr + g_{AB}(U^A du + dx^A)(U^B du + dx^B). \end{aligned} \quad (5)$$

This is pure notation, except for the choice $g_{ur} \equiv e^\beta$ which implies $\langle \partial_u, \partial_r \rangle < 0$, which in turn can be enforced within remnant coordinate freedom by appropriate choice of the direction in which u or r increases.

If we further choose⁵ the determinant of the induced metric on the cuts to scale like r^4

$$\partial_r \det \left(\frac{g_{AB}}{r^2} \right) = 0 \quad (6)$$

we say that r has been taken to be *areal* or *luminosity* distance. Then we write $g_{AB} = r^2 h_{AB}$ and

$$ds^2 = -Fdu^2 - 2e^\beta dudr + r^2 h_{AB}(U^A du + dx^A)(U^B du + dx^B). \quad (7)$$

Any coordinate system obeying $g_{rA} = g_{rr} = 0$ plus the luminosity distance condition is said to be in *Bondi gauge*; it is left to the reader's geometric imagination that every metric in this gauge can be cooked up by the geometric story we have told here.

The archetypal example of a spacetime described in Bondi gauge in $3+1$ Minkowski space where, starting from some inertial coordinates $(t, \vec{x})$, we take $r = |\vec{x}|$ to be spatial distance from the original, $u = t - r$ to be retarded time, and (x^A) to be any convenient set of coordinates on the celestial sphere. For example, taking $(x^A) = (\theta, \phi)$ we have

$$ds^2 = -du^2 - 2dudr + r^2(d\theta^2 + \sin^2 \theta d\phi^2). \quad (8)$$

In studying asymptotic symmetries and celestial holography it is frequently useful to replace the angular coordinates (θ, ϕ) with complex coordinates $(z, \bar{z})$ induced by, *e.g.*, stereographic projection.

⁵This cannot in general be accomplished by a redefinition of the coordinate r ; it constrains the possibilities for choosing $\{N_u\}$ at the start of this construction. For example, it is possible in flat Minkowski space if we take N_u to be the forward lightcone of the point $(t, x, y, z) = (u, 0, 0, 0)$ or equivalently the locus $t - r = u$ but not if we take N_u to be the locus $t - x = u$.